

Module 8. Deep Learning

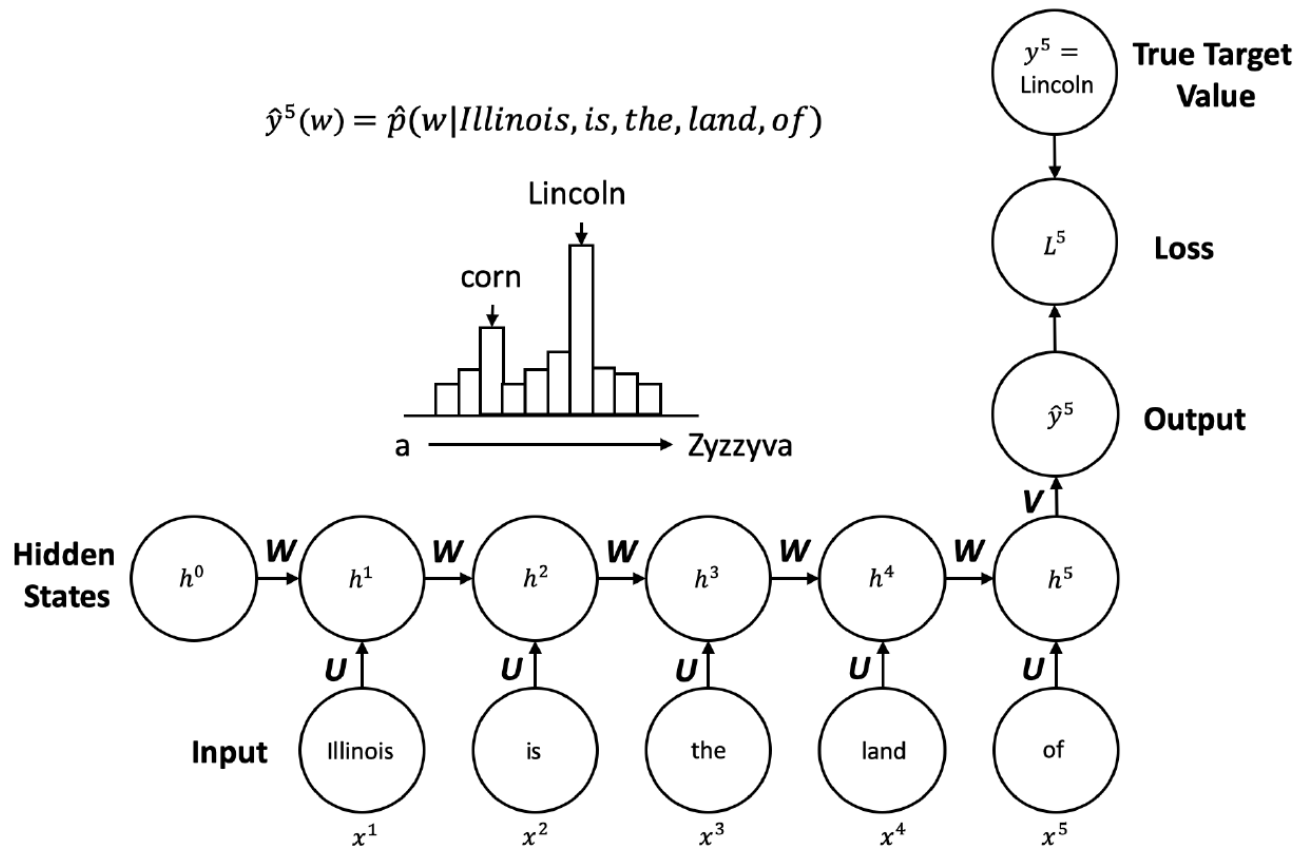
- ❑ Motivation
- ❑ Improve Training of Deep Learning Models
- ❑ Convolutional Neural Networks
- ❑ Recurrent Neural Networks 
- ❑ Summary

Recurrent Neural Networks

□ Handling sequences with **Recurrent Neural Networks (RNN)**

- At each time step, the input and the previous hidden state are fed into the network

$$h^t = f(Ux^t + Wh^{t-1} + a)$$
$$\hat{y}^T = g(Vh^T + b).$$



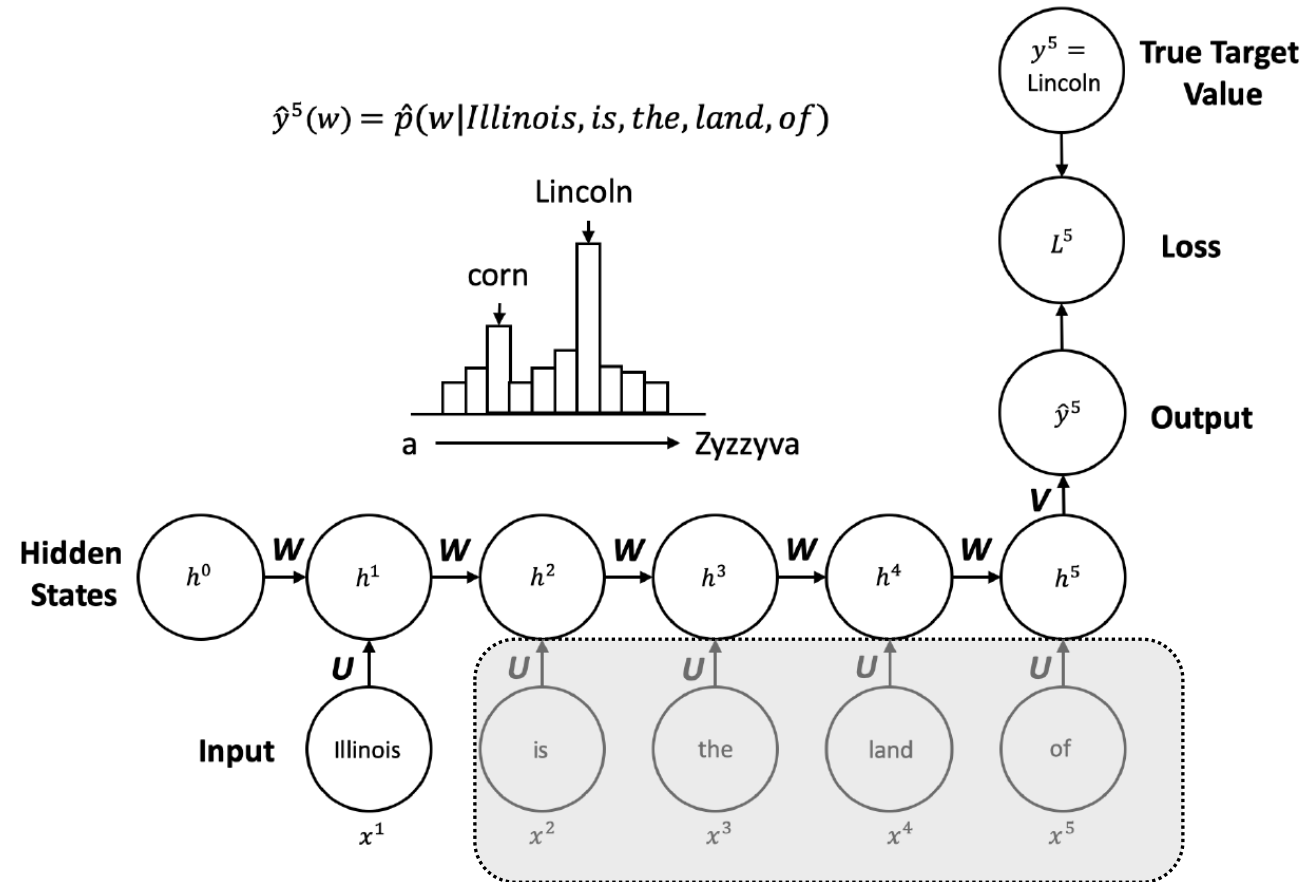
RNNs vs. Feed-forward NNs

□ RNNs:
$$h^t = f(Ux^t + Wh^{t-1} + b)$$
$$\hat{y}^T = g(Vh^T + b).$$

□ Feed-forward NNs:
$$h_i = f(W_i h_{i-1} + b_i)$$

□ Key Differences: Recurrence!

- same hidden-to-hidden weight matrix W
- same input-to-hidden weight matrix U
- same bias vector b



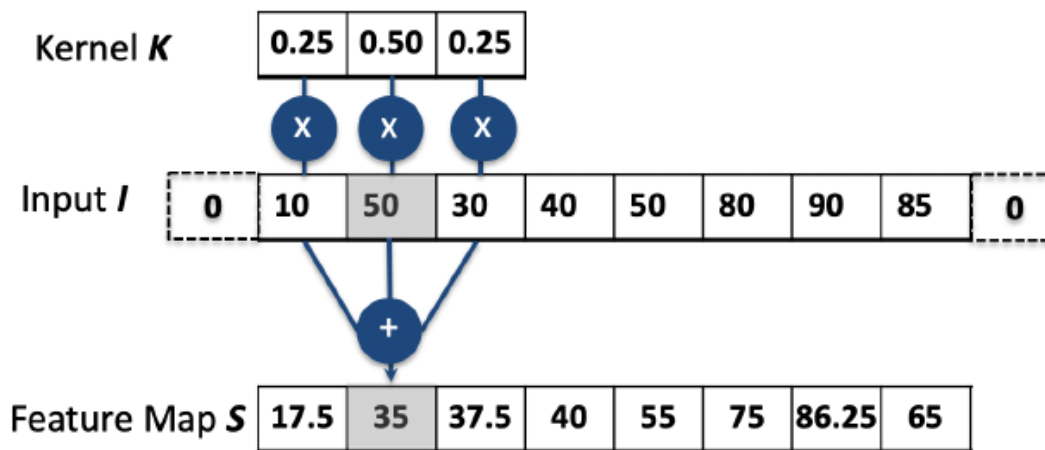
RNNs vs. CNNs

☐ Observations: sequence = 1D grid data

- ☐ Choice #1: RNNs
- ☐ Choice #2: 1D CNN

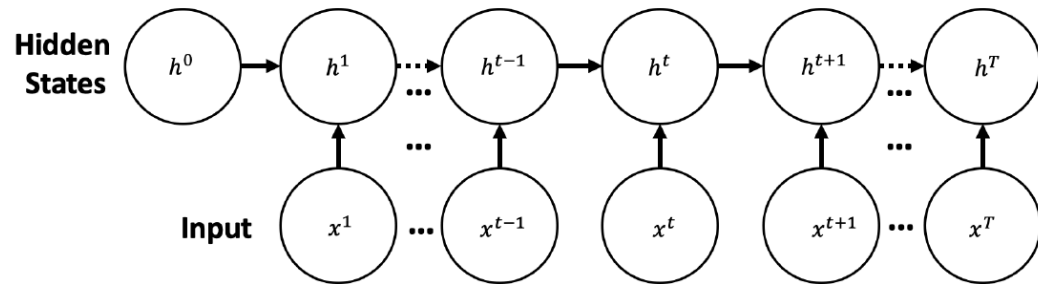
☐ Key Differences

- ☐ CNNs: localized (by convolution)
- ☐ RNNs: (in principle) capture long-term dependence (by recurrence)

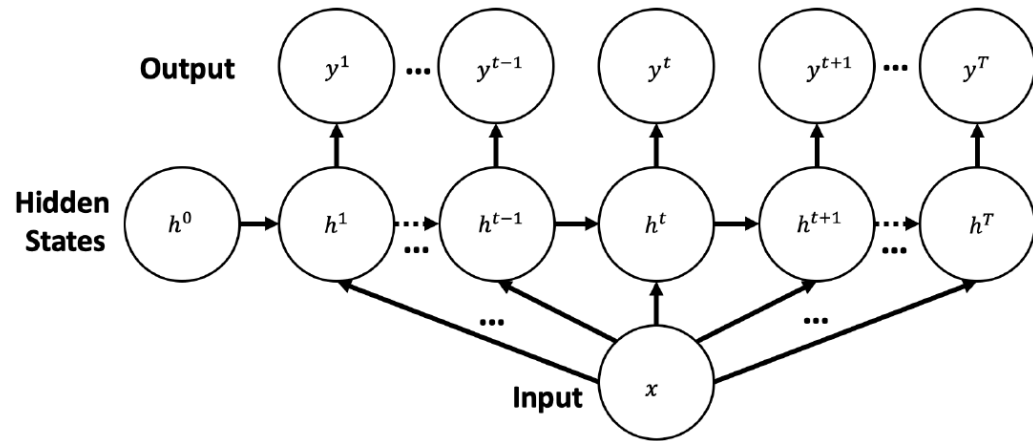


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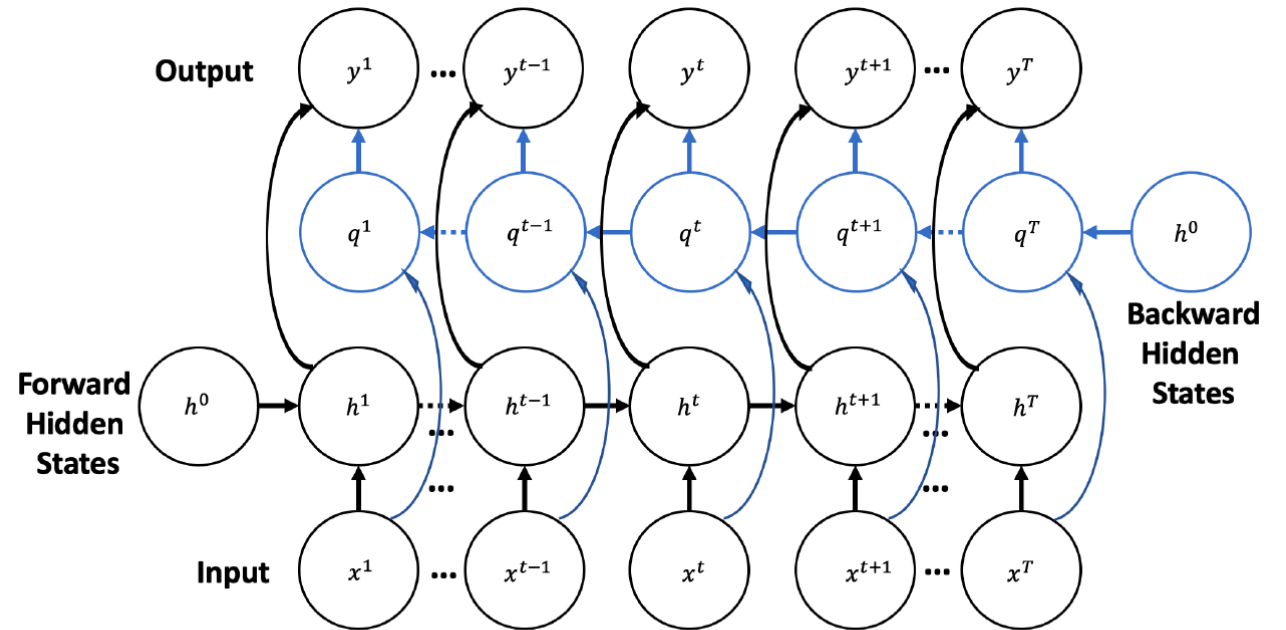
Different Types of RNNs



(a) RNN without output



(b) RNN with vector input



(c) bi-directional RNN